Alexander Bernstein

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Education

University of California, Santa Barbara — Santa Barbara, CA
Ph.D. in Statistics — *Sept. 2025*
Dissertation: "Long-Only Minimum Variance Portfolio Composition for Factor Models"

Washington University in St. Louis — St. Louis, MO
M.Sc. in Systems Science and Applied Mathematics — *Dec. 2016*
B.A. in Mathematics and Economics (Cum Laude) — *May 2014*

Technical Skills

Languages & Tools: Python (NumPy, SciPy, Pandas, Scikit-Learn, PyTorch, CVXPY), SQL, R, C, Java, JavaScript/NodeJS, FastAPI, Flask, REST APIs, Docker, Git, Linux, AWS, MongoDB, LaTeX.

Machine Learning: Neural Networks, Gaussian Processes, Reinforcement Learning (Q-Learning), Ensemble Methods (Boosting/Bagging), Bayesian Inference, Time Series Forecasting, Feature Engineering.

Data & Modeling: Statistical Modeling, Experimentation, High-Dimensional Regression, Optimization, End-to-End Pipelines, Model Deployment.

Selected Projects

End-to-End Equity Modeling Pipeline: Architected a modular, spec-driven (**YAML**) **Python** pipeline for portfolio construction from universe of S&P 500 assets: pluggable covariance/return estimators via decorator registries, **Numba**-JIT Kalman filters, **CVXPY** optimization, a walk-forward backtest engine, and a CLI rebalancer that emits live trades. Built on clean ABCs with unit tests and reproducible configs; runs the full **S&P 500** (daily/weekly data since 2010) in ∼5–10 min per strategy.

ML Surrogates for Derivative Pricing: Trained **Gaussian Process** (Matérn) and **PyTorch** neural-network surrogates to emulate Monte Carlo and finite-difference (Crank-Nicolson) solvers for the **Heston** and **CEV** models, recovering prices and **Greeks** (via autograd) at a fraction of the numerical cost.

Portfolio Aggregator (production web app): Built and self-hosted a **Dockerized FastAPI** dashboard that aggregates multi-brokerage accounts through a third-party **REST API** (SnapTrade) into a SQLite store, with scheduled auto-sync, historical snapshots, and daily-change tracking.

Wellbore Geology Prediction (Kaggle): Built the geologic-surface stage of a two-stage pipeline predicting subsurface layer depth along horizontal wells: **SPDE Matérn** Gaussian-random-field surfaces fit over nearest calibration wells with a multi-marker ensemble, composed with an HMM-style **Viterbi** dynamic-programming refinement and Bayesian smoothness-averaging. Multi-marker surfaces substantially improved local out-of-fold RMSE.

Experience

University of California, Santa Barbara — Santa Barbara, CA
Teaching Assistant & Graduate Student Mentor — *Sept. 2017 – June 2025*
Taught statistics, probability, regression, time series, and stochastic processes; mentored undergraduate research projects in modeling and optimization, including posters and written reports.

Epic Systems — Madison, WI
Technical Services Engineer — *Sept. 2014 – Sept. 2015*
Partnered with large institutional clients to deploy and customize the Epic EHR in production, diagnosing issues and shipping codebase improvements under high-availability and regulatory constraints.

Prozess Technologie — St. Louis, MO
Intern — *May 2013 – May 2014*
Built computational simulations to validate laboratory equipment and calibrated pharmaceutical manufacturing hardware to industry standards.

Research & Publications

Banerjee, T., **Bernstein, A.**, Feinstein, Z. (2025). "Dynamic clearing and contagion in financial networks." *European Journal of Operational Research* 321(2), 664–675.

Bernstein, A., Shkolnik, A. (2025). "Asymptotics of Quadratic Forms on a Simplex." *In preparation.*